Arithmetic Primitive-Orbit Rigidity in Stationary CCR Polynomials:
Shift Companions, Prime Divisors, and Bernoulli Ladders

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25 October 2026
Audit-informed revised research manuscript

Abstract

We investigate arithmetic root sharing in the Nörlund–Stirling shadows of formal stationary chronological-carry-refinement (CCR) polynomial transport. The central statements are established theorems, while an explicitly delimited stronger uniform congruence is stated as an open problem. First, trailing-source interpolation and a sharp obstruction to fixing all shadow-root positions separate controllability from rigidity. An exact shift-companion relation characterizes common factors of adjacent columns by the self-translate gcd of a single odd-order polynomial, yielding a sharp degree-pairing obstruction. Second, source-independent nonsharing is proved for sufficiently long irreducible Frobenius factors of four consecutive degrees, with exact symbolic Euclidean certificates in the difficult cases. Third, using Feit's large-Zsigmondy-prime theorem and an exact negative-coordinate evaluation, we establish for *every* odd order a rational irreducible factor absent from all predecessor columns. This implies polynomial noncoverage at all but finitely many good primes for each order and a positive-density family of primes with genuinely new linear residue roots. A coefficient-content criterion completely decides any fixed-order exceptional prime; exact certificates rule out all such primes for odd orders through 71. Finally, Bernoulli value identities and a triangular integer-coordinate recurrence produce contiguous blocks of target roots and explicit source-universal primitive p-adic residue disks. The universal *every-order, every-good-prime* nondivisibility open problem is *not* proved. Our contributions concern formal stationary polynomial transport, not physical pushdown-automaton realizability or the priority of the classical underlying number-theoretic ingredients.

Keywords: Nörlund polynomials; Stirling polynomial factors; primitive root orbits; large Zsigmondy primes; Bernoulli congruences; shift resultants; Hensel lifting; CCR transport.
2020 Mathematics Subject Classification: 11B68 (primary); 11S05, 12F10, 11A07 (secondary).

Contributions and novelty

The distinct contributions of this article, relative to the cited classical Nörlund–Stirling literature and the authorś earlier stationary CCR work, are as follows.

1. A source-uniform interpolation versus fixed-shadow rigidity dichotomy, including the precise upper limit on simultaneous anchoring of all split shadow-root positions.

2. The exact adjacent-column shift-companion identity, its multiplicity-sensitive polynomial-gcd formulation, and the resulting degree-pairing obstruction.

3. A family of primitive long-Frobenius-orbit criteria in four explicit factor degrees, and an all-odd-order rational predecessor noncoverage theorem derived using the cited large-Zsigmondy result.

4. An effective coefficient-content criterion for each fixed odd order, together with an exact finite, computer-assisted noncoverage certificate at every good prime for odd orders through 71.

5. A Bernoulli integer-coordinate ladder and explicit p-adic first-occurrence examples, with singular-root limitations kept separate from simple-root lifting.

The generating functions, classical number-theoretic theorems, and general computational-algebra tools are prior work. We do not claim an all-order, all-good-prime noncoverage proof or physical realization by a pushdown automaton.

1 Introduction and provenance

The Nörlund coefficients

$$S_j(X) = [t^j]\left(\frac{t}{1-e^{-t}}\right)^X \tag{1}$$

are classical polynomials. Their arithmetic normalization and their relation to Stirling coefficients are established topics; see Adelberg [5], Bencherif–Garici [6], McCarthy [10], and the NIST reference [7]. We make no claim to introducing this generating series or classical Hensel, resultant, subresultant, or Chebotarev methods. The stationary chronological-carry-refinement (CCR) recurrence and its source interpretation originate in the author's earlier CCR-3, CCR-4, CCR-k, and arithmetic-refinement studies [1, 2, 3, 4]. This article instead concerns whether an entire Frobenius orbit of a later column can have occurred in an earlier column, *uniformly across stationary source choices*.

The contribution combines several distinct, logically separated pieces. An exact *shift-companion characterization* converts an adjacent-column common factor into a self-translate common factor of one polynomial; it is proved without a finite calculation. A range of *large-factor primitive-orbit criteria* follows from a small number of leading coefficients together with exact Euclidean certificates for the harder codimensions. These tools complement, rather than imply, the global rational and Bernoulli congruence arguments developed below. A finite predecessor-product certificate through order 71 is reported separately; its finite range must not be confused with the uniform irreducible-orbit theorems.

2 The stationary model and good-prime normalization

For a prime p and a source vector $\boldsymbol{\lambda} = (\lambda_3, \lambda_4, \ldots)$, define $F_0 = 1$, $F_j = 0$ for $j < 0$, and

$$\begin{aligned} T_jF_j(X) &= ((p+1)X - p)F_{j-1}(X) \\ &\quad + p((p+1)(X-1) - p)F_{j-2}(X-1) \\ &\quad + p\sum_{s=3}^{j}(-1)^s\lambda_s F_{j-s}(X-s+1), \\ T_jh(X) &= Xh(X+1) - (X-j)h(X). \end{aligned} \tag{2}$$

The diagonal of T_j on polynomials of degree at most j is $j, j+1, \ldots, 2j$. Fix an integer N and assume throughout a *good prime* $p > 2N$. Then every inverse T_j^{-1}, $j \le N$, is p-integral, and

$$F_j(X; p, \boldsymbol{\lambda}) \equiv S_j(X) \pmod{p} \qquad (j \le N,\ \lambda_s \in \mathbb{Z}_p). \tag{3}$$

This is coefficientwise congruence. No statement here is made about physical machines.

For odd $n \geq 5$ the shadow has the rational factorization

$$S_n(X) = \frac{X^2(X-1)}{2^n n!}\, f_n(X), \qquad f_n \in \mathbb{Q}[X] \text{ monic}, \quad \deg f_n = n-3. \tag{4}$$

The polynomial f_n is the monic normalization of the primitive integral Stirling quotient q_n used in the CCR literature. The normalizing leading coefficients are p-units in the good-prime range. We always use gcd to mean the *monic* polynomial greatest common divisor over the relevant field.

An irreducible monic factor $g \mid \overline{f}_n$ over $\mathbb{F}_p$ is called *primitive relative to the preceding columns* when

$$\gcd(g, \overline{S_m/X}) = 1 \qquad (1 \leq m < n). \tag{5}$$

A bar indicates coefficientwise reduction. The condition is unaffected by the choice of a common integral source vector because of (3). For $\deg g \geq 2$ it is equivalent to nonsharing against the full earlier shadows S_m.

3 Source interpolation and positional rigidity

A source-vector obstruction and a root-location obstruction are not the same. Even when the shadow factorization is fixed, the stationary source parameters can be used to move the lifted roots. The following two theorems give a sharp control/rigidity distinction; neither should be read as a statement about the number of roots of the deformed target polynomial.

Theorem 3.1 (Universal interpolation by a trailing source block)**.** *Fix $n \geq 5$, a prime $p > 2n$, and $1 \leq d \leq n-2$. Fix arbitrary sources $\lambda_3, \ldots, \lambda_{n-d} \in \mathbb{Z}_p$ (when present). Let $a_1, \ldots, a_d$ be pairwise distinct zeros of $\overline{S}_n$ and let $\alpha_i \in \mathbb{Z}_p$ satisfy $\alpha_i \equiv a_i \pmod p$. There exists a* unique *tail tuple $(\lambda_{n-d+1}, \ldots, \lambda_n) \in \mathbb{Z}_p^d$ for which*

$$F_n(\alpha_1) = \cdots = F_n(\alpha_d) = 0.$$

The a_i need not be simple target-shadow roots. If the prescribed points are roots of fixed earlier columns of order at most $n-d$, those parent columns remain unchanged.

Proof. Set $C_{n,s}(X) = p^{-1}\partial F_n/\partial\lambda_s \pmod p$. Differentiation of the stationary recurrence, or extraction of the linear source contribution from its generating series, gives

$$C_{n,s}(X) = (-1)^s [t^n] A(t)^X \int_0^t z^{s-1} A(z)^{-(s-1)}\, dz \pmod p.$$

Since $A(t) = 1 + t/2 + O(t^2)$, the polynomial $C_{n,s}$ has degree exactly $n-s$, with leading coefficient

$$[X^{n-s}] C_{n,s} = \frac{(-1)^s}{s\, 2^{n-s}(n-s)!} \quad \text{in } \mathbb{F}_p^\times.$$

Thus $C_{n,n}, C_{n,n-1}, \ldots, C_{n,n-d+1}$ form a polynomial basis of degrees $0, 1, \ldots, d-1$. Their evaluation matrix at the distinct a_i has a nonzero Vandermonde determinant, multiplied by nonzero leading coefficients. Because each $\overline{S}_n(a_i)$ vanishes, the vector map $(F_n(\alpha_i)/p)_i$ is integral and affine modulo p in the tail variables, with exactly this invertible Jacobian. Nonlinear source terms contribute at least p^2 before the division by p. The multivariate Hensel lemma produces the unique $\mathbb{Z}_p$ solution, to all source digits. A tail source of order above a parent column cannot modify that parent. □

A complementary source-independent coefficient survives *all* stationary source deformations. Put $R_n = S_n/X$, and in the zero-source transport allow a formal transport parameter P in place of the prime p; denote $E_n = \partial_P F_n(X; P, 0)|_{P=0}$.

Lemma 3.2 (The uncorrectable transport drift)**.** *For every odd $n \geq 5$,*

$$\boxed{[X^{n-2}] \operatorname{rem}_{R_n} E_n = -\frac{4(n+3)}{9\,2^n(n-2)!}.} \tag{6}$$

Moreover, every source-dependent summand of F_n has X-degree at most $n-3$, even when several sources interact nonlinearly.

Proof. The second statement follows by induction in (2): the source insertion at order $s \geq 3$ starts at degree at most $n-s$, and T_n^{-1} preserves the degree filtration. For the first statement, extract the three highest X-coefficients from the zero-source generating series $H_P(t)\exp(Xf_P(t))$ and its derivative at $P=0$; the exact series expansion at $P=0$ gives $f_0(t) = t/2 - t^2/24 + O(t^4)$ and derivative coefficients $a' = 1/2, b' = 1/4, c' = -1/6$. The polynomial division by R_n has quotient $nX + n(4n-10)/3$ at its top two coefficients. Substituting the extracted coefficients yields (6). The equality is obtained by symbolic coefficient extraction and polynomial division, not by extrapolation from finitely many numerical orders. Its top coefficient is the displayed nonzero rational function of n. □

Theorem 3.3 (Sharp fixed-shadow anchor capacity)**.** *Let $n \geq 5$ be odd, $p > 2n$, and suppose $R_n \bmod p$ splits into $n-1$ distinct roots, with exact p-adic shadow-root positions $\beta_1, \ldots, \beta_{n-1}$. Then any selected $n-2$ of these positions can be made simultaneous zeros of F_n by a unique integral source vector $\lambda_3, \ldots, \lambda_n$, but* all *$n-1$* positions cannot *be made zeros even modulo p^2. When $n-2$ are anchored, the evaluation at the remaining exact shadow position has p-adic valuation exactly one.*

Proof. Existence and uniqueness follow from Theorem 3.1 with $d = n-2$. For impossibility, reduce $p^{-1}(F_n - S_n)$ modulo p and divide it by R_n. The source contribution has degree at most $n-3$, whereas Lemma 3.2 shows the remainder has exact degree $n-2$ and p-unit leading coefficient, independently of all sources. It cannot vanish at $n-1$ distinct residue classes. If it vanishes at any $n-2$ classes, it is a unit multiple of their monic root polynomial; its value in the last residue class is a unit, so the corresponding evaluation of F_n has valuation exactly one. □

Corollary 3.4 (Source-independent weighted first motion)**.** *Retain the complete splitting hypothesis of Theorem 3.3. Write $a_i = \overline{\beta}_i$ and allow moving targets $\gamma_i = \beta_i + p\delta_i$. If all $F_n(\gamma_i) \equiv 0 \pmod{p^2}$, then*

$$\boxed{\sum_{i=1}^{n-1} a_i \delta_i \equiv \frac{4n(n-1)(n+3)}{9} \pmod{p}.} \tag{7}$$

Proof. For $H = (F_n - S_n)/p \pmod{p}$, its remainder modulo R_n has top coefficient κ_n from (6). Lagrange interpolation on the distinct β_i gives $\sum_i H(\beta_i)/R_n'(\beta_i) = \kappa_n/L_n$, where $L_n = 1/(2^n n!)$ is the leading coefficient of R_n. Because $S_n = XR_n$, Taylor expansion at $\gamma_i = \beta_i + p\delta_i$ gives $F_n(\gamma_i)/p \equiv H(\beta_i) + a_i\delta_i R_n'(\beta_i) \pmod{p}$. Sum after dividing by $R_n'(\beta_i)$ and insert $\kappa_n/L_n = -4n(n-1)(n+3)/9$. □

The fixed-position capacity concerns preassigned exact shadow zeros, not freely moving roots; in particular it does not contradict the source-universal primitive residue-orbit results below. It also does not imply a no-go for every possible assignment of roots from earlier *moving* CCR columns, which can depend on some of the same tail sources.

4 The exact shift-companion identity

Theorem 4.1 (Shift-companion theorem)**.** *For every odd $n \geq 5$ the following identities hold over $\mathbb{Q}[X]$:*

$$\frac{S_{n-1}(X)}{X} = \frac{(X+1)^2 f_n(X+1) - (X-n)(X-1)f_n(X)}{2^n n!}, \tag{8}$$

$$f_n(-1) = \frac{2^{n-1}}{n+1}. \tag{9}$$

Consequently, for each prime $p > 2n$,

$$\boxed{\gcd\left(\overline{f}_n, \overline{S_{n-1}/X}\right) = \gcd\left(\overline{f}_n(X), \overline{f}_n(X+1)\right).} \tag{10}$$

The equality is valid with multiplicities and does not require $\overline{f}_n$ to be squarefree.

Proof. The shadow recurrence obtained from (1) is $T_n S_n(X) = X S_{n-1}(X)$. Substitute (4), expand T_n, and divide by X twice to obtain (8). At $X = -1$, direct coefficient extraction yields $S_n(-1) = [t^n](1 - e^{-t})/t = (-1)^n/(n+1)! = -1/(n+1)!$. Using (4) at -1 gives (9). Because $p > 2n$, $(n+1)2^n n!$ is invertible modulo p, so $\overline{f}_n(-1)$ is nonzero. Reducing (8) modulo $\overline{f}_n$ gives $\overline{S_{n-1}/X} \equiv (X+1)^2 \overline{f}_n(X+1)/(2^n n!)$. The factor $(X+1)^2$ is relatively prime to $\overline{f}_n$ because $\overline{f}_n(-1) \neq 0$. Taking monic gcds proves (10). □

Corollary 4.2 (Degree pairing of adjacent-column Frobenius orbits)**.** *Suppose $p > 2n$ and $g \mid \overline{f}_n$ is irreducible of degree k and also divides $\overline{S_{n-1}/X}$. Then the distinct irreducible translate $g(X-1)$ also divides $\overline{f}_n$. Therefore*

$$\boxed{2k \leq n - 3.} \tag{11}$$

In particular, every irreducible factor of $\overline{f}_n$ with degree greater than $(n-3)/2$ is absent from the immediate *preceding shadow column.*

Proof. By (10), $g(X)$ divides $f_n(X+1)$; equivalently $g(X-1)$ divides $f_n(X)$. The polynomials $g(X)$ and $g(X-1)$ cannot coincide: for a monic degree-k polynomial their X^{k-1} coefficients differ by $-k$, which is nonzero because $1 \leq k \leq n-3 < p$. Thus two distinct monic irreducible factors, each of degree k, divide $\overline{f}_n$; their product divides it, giving $2k \leq \deg f_n = n - 3$. □

Corollary 4.3 (Exact exceptional-prime test)**.** *If a good prime supports any common factor of $\overline{f}_n$ and $\overline{S_{n-1}/X}$, it divides the numerator of the reduced rational resultant*

$$\mathrm{Res}_X(f_n(X), f_n(X+1)). \tag{12}$$

When this resultant is a nonzero rational number, the converse holds away from its denominators: its good-prime divisors are exactly the adjacent-column common-factor primes.

Proof. The statement is the standard resultant criterion applied to (10); good primes do not divide any leading coefficient or denominator used in normalization. □

For $(n, p) = (9, 23)$, the consecutive quotient roots are $19, 20 \pmod{23}$; the common root with order 8 is 19. Table 1 is a small exact illustration, not a claim about the factorization of all self-shift resultants. The general identities above do not depend on the table.

Table 1: Exact self-shift resultant prime support for primitive integral q_n (not for monic f_n).

n	$\lvert\operatorname{Res}(q_n(X), q_n(X+1))\rvert$	eligible good-prime support
5	$2^6 3^2$	none
7	$2^{27} 3^9$	none
9	$2^{54} 3^{15} 5^4 \cdot 23$	23

5 Primitive long Frobenius factors at every odd order

The next theorem organizes the uniform long-orbit statements obtained from the leading coefficients of the shadows. The first two cases admit short direct coefficient proofs; the last two are supported by symbolic Euclidean certificates whose exceptional branches are described below.

Theorem 5.1 (Prime-indexed primitive long-orbit theorem)**.** *Let $n \geq 5$ be odd, $p > 2n$, and $d = n - 3$. Suppose $\overline{f}_n$ has an irreducible factor g of one of the following degrees, within the stated range:*

$\deg g$	*valid orders*
d	$n \geq 5$
$d-1$	$n \geq 7$
$d-2$	$n \geq 9$
$d-3$	$n \geq 11$.

(13)

Then g is primitive relative to every *earlier shadow column:* $\gcd(g, \overline{S_m/X}) = 1$ *for* $1 \leq m < n$.

Proof and symbolic certificate. Put $L_j = 1/(2^j j!)$ and write the monic top coefficients of S_j as

$$\frac{S_j(X)}{L_j} = X^j + a_j X^{j-1} + b_j X^{j-2} + \cdots, \quad a_j = -\frac{j(j-1)}{6}, \quad b_j = \frac{j(j-1)(j-2)(j-3)}{72}. \tag{14}$$

For $Q = f_n$ (monic) the first coefficients are

$$Q = X^d + AX^{d-1} + BX^{d-2} + \cdots, \quad A = 1 - \frac{n(n-1)}{6}, \quad B = A + \frac{n(n-1)(n-2)(n-3)}{72}. \tag{15}$$

The case $\deg g = d$ follows by comparing degrees against each S_m/X. At $m = n-2$ the next-to-leading discrepancy is $2(n-3)/3$; at $m = n-1$ the forced linear quotient leaves discrepancy $2(n-3)(n-1)/9$. Both are units in $\mathbb{F}_p$. This is also a direct proof that Q cannot divide a *single* earlier column.

For $\deg g = d-1$ only $m = n-3, n-2, n-1$ require attention; smaller orders have insufficient degree. The $m = n-3$ forced linear quotient has top remainder $-2(n-3)/3$, a unit. At $m = n-2$ the odd-column quotient factorization reduces the potentially shared long factor to one of degree $d-2$, too small. At $m = n-1$ the top remainder of S_{n-1}/X upon division by Q has degree $d-2$ and leading coefficient $2(n-3)(n-1)/9$, excluding a shared degree-$(d-1)$ factor.

For the remaining degrees we use *exact algebraic certificates in the variable n*, not sampling over a bounded set of n. The coefficient-extraction input, which is reproducible from (1), is

$$\sigma_k(j) = 2^k (j)_k [z^k] \left(1 - \frac{z}{12} + \frac{z^3}{1440} - \frac{z^5}{90720} + \frac{z^7}{4838400} - \frac{z^9}{239500800} + \cdots\right)^{j-k}. \tag{16}$$

Here $(j)_k$ is the falling factorial, and only terms of degree at most k are used. The monic coefficients of Q are $Q_k(n) = \sum_{i=0}^{k} \sigma_i(n)$. In any quotient-by-predecessor calculation, the monic coefficients of S_m/X are $\sigma_k(m)$. We apply the ordinary Euclidean algorithm symbolically, splitting off any vanishing-leading-coefficient cases *before* normalizing a remainder.

For $\deg g = d-2$ and $d-3$, the possible predecessor orders are confined to $m \geq n-4$ and $m \geq n-5$, respectively, by degree and the odd predecessor's known factors $X(X-1)q_m$. The symbolic reductions and their prime filters are summarized in Table 2. For the $m = n-5$ case, the two necessary remainder numerators (after removing factors that are units in the good-prime range) are

$$n^2 + 3n + 18, \qquad 11n^3 + 11n^2 + 492n - 3132.$$

Their exact integer resultant is $2^{11}3^4 \cdot 7 \cdot 11$, leaving no prime $p > 2n$ in this branch. For $m = n-2$, the corresponding necessary numerators are

$$27n - 56, \qquad 405n^2 - 2330n + 2064,$$

whose resultant is $-2^6 3^3 \cdot 433$. At $p = 433$, the required residue is $n \equiv 403$, incompatible with $2n < p$. These are exact polynomial resultant identities, not conclusions from a bounded scan. For the degree-$(d-2)$ case, use the *corrected* last-predecessor Euclidean chain from the degree-$(d-3)$ certificate; an older quadratic-quotient version is superseded.

Table 2: Degree-$(d-3)$ common-factor exclusions, $n \geq 11$ odd. An *eligible* branch means $p > 2n$ and all division leading coefficients are units.

Predecessor	Exact obstruction/certified elimination
$m \leq n-6$	$\deg(S_m/X) = m-1 < d-3$.
$m = n-4$ (odd)	S_m/X factors as $X(X-1)q_m$, with $\deg q_m = n-7 < d-3$.
$m = n-5$	Two cubic-division conditions have resultant supported only on $2, 3, 7, 11$.
$m = n-2$	Two successive subresultant conditions yield $p = 433$ only; the forced residue $n \equiv 403 \pmod{433}$ violates $p > 2n$.
$m = n-3$	Three Euclid conditions leave $23, 1063, 3862141$; the admissible candidates are eliminated by parity, no common residue, or an earlier degree drop.
$m = n-1$	The correctly indexed chain has unit $r_0 = 2(n-3)(n-1)/9$ and unit $u_0 = -(n-2)(3n-11)/75$. The last conditions leave $151, 4967, 236479$, eliminated by parity, no common residue, or degree drop.

All certificates are polynomial identities in n and integer resultant factorizations. At the two exceptional branches where the next leading coefficient vanishes, the *degree drops below the proposed irreducible factor degree*, rather than creating an additional solution. Thus no good-prime pair (n, p) survives in any of the predecessor classes. For $m = n-3$, the reduced resultant-prime support is $\{2, 3, 5, 7, 23, 1063, 3862141\}$; at 23 the only common residue is even, at 1063 there is no common residue, and the residual candidate at 3862141 makes the next leading coefficient vanish, dropping the gcd degree. For $m = n-1$, the corresponding support is $\{2, 3, 5, 7, 151, 4967, 236479\}$; the candidates are excluded, respectively, by parity, absence of a common root, or a degree drop. These latter resultant supports and exceptional-residue eliminations are recorded exact symbolic certificate outcomes, not independently reconstructed from every coefficient in the present revision. The proof remains computer-assisted at those steps and does not infer the all-order result from a finite scan. □

Corollary 5.2 (Source-universal unramified factors)**.** *Under the hypotheses of Theorem 5.1, for every common stationary source vector $\boldsymbol{\lambda} \in \mathbb{Z}_p^{n-2}$ the target polynomial F_n has a unique monic p-integral factor $G_{n,g}$ reducing to g. It is irreducible over $\mathbb{Q}_p$, defines the unramified extension of degree* $\deg g$*, and shares no root with any F_m, $m < n$, under the* same *source vector.*

Proof. An irreducible factor of a finite-field polynomial is separable; in the four degree ranges of Theorem 5.1 the residual total degree $d - \deg g \leq 3$ is strictly smaller than $\deg g$ except at the small excluded orders, so g occurs with multiplicity one. The remaining trivial factors

$X^2(X-1)$ are relatively prime to g because $\deg g \geq 2$. Apply Hensel's coprime-factor lifting over $\mathbb{Z}_p$ to (3). The lifted factor defines an unramified extension, and any shared root with earlier F_m would reduce to a forbidden common residue root of g and S_m. □

6 Every odd order has a new rational irreducible factor

Write $\mathcal{B}_n = \prod_{m=2}^{n-1} P_m/X$ for the primitive integral predecessor product, where P_m denotes the primitive positive-leading integral numerator of S_m. The distinction between a *rational factor missing from every predecessor* and the stronger *absence of all possible mod-p coverage* is essential.

Lemma 6.1 (Negative integer specialization)**.** *For integers $a, n \geq 1$,*

$$S_n(-a) = \frac{(-1)^n a!}{(n+a)!} \left\{ {n+a \atop a} \right\}, \quad \textit{in particular} \quad S_n(-2) = \frac{2(-1)^n(2^{n+1}-1)}{(n+2)!}. \tag{17}$$

Proof. Since $A(t)^{-a} = t^{-a}(1-e^{-t})^a$, apply the binomial theorem and the ordinary finite-difference representation of the Stirling numbers of the second kind. The special case $a = 2$ then follows from $\left\{ {n+2 \atop 2} \right\} = 2^{n+1} - 1$. □

Lemma 6.2 (Boundary-prime integral shadows)**.** *If $r \geq n+2$ is prime, all coefficients of S_j, $0 \leq j \leq n$, belong to $\mathbb{Z}_{(r)}$, and their leading coefficients $1/(2^j j!)$ are units there. In particular, primitive normalization and polynomial evaluation commute with reduction modulo r.*

Proof. The coefficients of $(1-e^{-t})/t$ through t^n have denominators dividing $(n+1)!$. Its formal inverse $A(t)$ is therefore integral at r. Expand $A(t)^X = (1+(A(t)-1))^X$ in binomial polynomials $\binom{X}{k}$, whose denominators $k!$ are r-units for $k \leq n$. The claimed leading coefficient is obtained from $A(t) = 1 + t/2 + O(t^2)$. □

Theorem 6.3 (Global rational predecessor noncoverage)**.** *For every odd integer $n \geq 5$, the polynomial $q_n \in \mathbb{Z}[X]$ has an irreducible factor $g_n \in \mathbb{Q}[X]$ that does* not *divide $\mathcal{B}_n$. In particular,*

$$\boxed{\operatorname{rad}(q_n) \nmid \mathcal{B}_n \quad \textit{and} \quad q_n \nmid \mathcal{B}_n \quad \textit{over } \mathbb{Q}[X].} \tag{18}$$

Proof. The decisive external input is Feit's *large Zsigmondy prime theorem* in the form verified by Roitman [8, 9]. For any $m > 2$ outside $\{4, 6, 10, 12, 18\}$ there exists a prime r such that $\operatorname{ord}_r(2) = m$ and either $r > m+1$ or $r^2 \mid 2^m - 1$. Put $m = n+1$ (even). If $r > m+1 = n+2$, the congruence $r \equiv 1 \pmod{m}$ gives $r \geq 2m+1 > 2n$. By Lemma 6.1, $S_n(-2) = 0 \pmod{r}$, while $S_j(-2) \neq 0 \pmod{r}$ for every $1 \leq j < n$ since $\operatorname{ord}_r(2) = n+1$ and all relevant denominators are r-units.

If the large divisor is instead $r = m+1 = n+2$ and $r^2 \mid 2^m - 1$, the denominator $(n+2)!$ in (17) has r-valuation exactly one, and the numerator has valuation at least two. Thus $S_n(-2) \equiv 0 \pmod{r}$ once again. For all $j < n$ the denominator $(j+2)!$ is an r-unit and the multiplicative order prevents a zero. Lemma 6.2 legitimizes every reduction at this borderline prime. This step is a *shadow-polynomial statement only*: the stationary recurrence need not admit source-independent reduction at $r \leq 2n$.

The four excluded even exponents $m = 6, 10, 12, 18$ correspond to target orders $n = 5, 9, 11, 17$. Their exact witnesses are, respectively, $(a, r) = (3, 23), (3, 19), (4, 29), (4, 3169)$ at the coordinate $-a$. Equation (17) and independent integer Stirling recurrences certify target vanishing and all preceding nonvanishing; the values are tabulated and reproducible in the supplement. In every case we have a prime $r \geq n+2$ and $a \in \mathbb{F}_r$ with

$$\overline{q}_n(a) = 0, \qquad \overline{\mathcal{B}}_n(a) \neq 0. \tag{19}$$

Factor the primitive integral q_n into primitive irreducible factors over $\mathbb{Z}[X]$. Because the leading coefficient of q_n is an r-unit, at least one factor g_n vanishes at a modulo r. If that factor divided $\mathcal{B}_n$ over $\mathbb{Q}[X]$, Gauss's lemma would give integral divisibility up to a unit and therefore $\overline{\mathcal{B}}_n(a) = 0$, contradicting (19). Thus g_n is missing from the predecessor product, proving the asserted radical nondivisibility. □

Corollary 6.4 (Cofinite good-prime noncoverage)**.** *For every fixed odd $n \geq 5$, one has $q_n \nmid \mathcal{B}_n$ over $\mathbb{F}_p[X]$ for all but finitely many primes $p > 2n$.*

Proof. Divide $\mathcal{B}_n$ by q_n over $\mathbb{Q}[X]$ and reduce the nonzero remainder from Theorem 6.3 modulo p. Only primes dividing a coefficient denominator, a leading coefficient or the content of its nonzero integer numerator can make the remainder vanish. □

Corollary 6.5 (A positive-density class with actual new residues)**.** *For each fixed odd $n \geq 5$, the set of primes $p > 2n$ admitting $a_p \in \mathbb{F}_p$ with $\overline{q}_n(a_p) = 0$ and $\overline{\mathcal{B}}_n(a_p) \neq 0$ has natural density at least $1/(n-3)!$.*

Proof. Choose the missing irreducible factor g_n of Theorem 6.3. Its resultant with $\mathcal{B}_n$ is nonzero over $\mathbb{Z}$, hence their reductions are coprime outside finitely many primes. The rational splitting field K of g_n has degree at most $(\deg g_n)! \leq (n-3)!$. By the Chebotarev density theorem [11], the primes splitting completely in K have density $1/[K : \mathbb{Q}]$. At every such prime away from the finitely many exclusions, g_n has a linear root absent from all predecessor polynomials. □

Remark 6.6 (Why this does not settle the universal congruence)**.** *For a fixed order, the nonzero rational remainder may vanish modulo finitely many good primes. Our results do not exclude* all *such exceptional primes, and the auxiliary boundary primes used in the proof of Theorem 6.3 are not necessarily good CCR transport primes. The density conclusion concerns linear* finite-field *roots; exact $\mathbb{Q}_p$-rational lifts for every source choice require a separate simple-root hypothesis.*

6.1 An explicit infinite family independent of factorization

A particularly transparent subfamily needs no large-Zsigmondy classification. For every $t \geq 3$ let $n = 2^t - 1$ and let p be any prime divisor of the Fermat number $2^{2^{t-1}} + 1$. Then $\mathrm{ord}_p(2) = 2^t = n+1$, and $p \equiv 1 \pmod{2^{t+1}}$, so $p > 2n$. Equation (17) gives $q_n(-2) = 0$ while $\mathcal{B}_n(-2) \neq 0$ modulo p. Distinct Fermat numbers are pairwise coprime, so this yields infinitely many *distinct* good primes and target orders, with a new source-independent residue root -2. No Fermat number is assumed prime.

7 Chebotarev densities from long irreducible factors

Assume for a fixed odd n that $\mathrm{Gal}(f_n/\mathbb{Q}) = S_d$. At an unramified prime, a degree-k irreducible factor corresponds to a k-cycle in Frobenius. For $k \in \{d, d-1, d-2, d-3\}$ and $d \geq 8$, the types are disjoint: two such long cycles cannot coexist. The proportion of permutations containing a k-cycle is $1/k$, because the remaining $d - k < k$ points may be permuted freely. Consequently, the primes on which Theorem 5.1 guarantees a source-universal, previously unseen unramified orbit have density *at least*

$$\frac{1}{d} + \frac{1}{d-1} + \frac{1}{d-2} + \frac{1}{d-3} \quad (d \geq 8), \tag{20}$$

subject to the stated verified Galois hypothesis. The bounds for the first four columns with established symmetric groups are:

n	d	$\mathrm{Gal}(f_n/\mathbb{Q})$	sufficient-prime density
5	2	S_2	1/2
7	4	S_4	7/12
9	6	S_6	37/60
11	8	S_8	533/840.

(21)

For $n = 5, 7, 9$ only the available degree cases are summed. These are *lower densities for a sufficient class*, not the density of every prime with a primitive orbit; no S_d hypothesis is assumed at orders where it has not been audited.

8 Exact content obstructions and an unconditional finite range

Proposition 8.1 (The fixed-order coefficient-content criterion)**.** *For odd $n \geq 5$, define*

$$\begin{aligned} H_n(X) &= \mathrm{rem}_{\mathbb{Q}[X]}(\mathcal{B}_n, q_n), \\ D_n &= \mathrm{lcm}\{\textit{coefficient denominators of } H_n\}, \\ C_n &= \mathrm{cont}(D_n H_n). \end{aligned}$$

Here $D_n > 0$ is minimal, the content is the positive gcd of the integer coefficients, and $C_n = 0$ if $H_n = 0$. Then for every prime $p > 2n$,

$$\boxed{\overline{q}_n \mid \overline{\mathcal{B}}_n \iff p \mid C_n.} \tag{22}$$

Moreover, Theorem 6.3 shows $C_n > 0$ for every *odd $n \geq 5$.*

Proof. The same p-integrality argument as in Lemma 6.2 (now $p > 2n$) shows the leading coefficient of q_n is a p-unit, and all coefficients of $\mathcal{B}_n$ and the quotient and remainder of Euclidean division lie in $\mathbb{Z}_{(p)}$. Reducing that division modulo p therefore gives the correct polynomial remainder over $\mathbb{F}_p$. It vanishes exactly when every coefficient of the integer polynomial $D_n H_n$ vanishes, or equivalently p divides its content. By Theorem 6.3, $H_n \neq 0$, so C_n is positive. □

The content criterion is an elementary *exact decision reduction*, rather than a claim that the content has a known closed form. Computation of H_n uses exact rational polynomial arithmetic and requires no search over primes. In a finite but nontrivial order range it produces a complete unconditional certificate.

Theorem 8.2 (All-good-prime noncoverage through order 71)**.** *For every odd $5 \leq n \leq 71$ and* every *prime $p > 2n$,*

$$\boxed{\overline{q}_n \nmid \overline{\mathcal{B}}_n \quad \textit{in } \mathbb{F}_p[X].} \tag{23}$$

Proof. The exact finite coefficient-content calculation gives $C_n = 2^{e_n}$ in all 34 odd orders; its complete exponent table is displayed below. One exact construction, using the Bernoulli numbers B_{2r}, is the recurrence

$$jS_j(X) = X\left(\frac{1}{2}S_{j-1}(X) - \sum_{r=1}^{\lfloor j/2 \rfloor} \frac{\mathsf{B}_{2r}}{(2r)!} S_{j-2r}(X)\right), \qquad S_0 = 1.$$

Construct the primitive positive-leading integral numerators P_m, form q_n, and successively take the exact rational remainder of $\prod_{m=2}^{n-1}(P_m/X)$ modulo q_n. The least coefficient-denominator multiple D_n and the gcd of the resulting integral coefficients yield C_n. The full recorded

certificate covers all 34 rows and cross-checks shadow construction by an independent transport recurrence. During this revision an independent exact-rational implementation reproduced the displayed contents through $n = 61$; the last five rows $(63, 65, 67, 69, 71)$ retain their earlier full-certificate provenance rather than being claimed as newly reexecuted. By Proposition 8.1, each recorded identity $C_n = 2^{e_n}$ excludes every prime $p > 2n$. The theorem is a finite computer-assisted result, not an induction over arbitrary odd n. □

Table 3: Exact integer-content exponents $C_n = 2^{e_n}$ for the 34 certified odd orders.

n	e_n	n	e_n	n	e_n	n	e_n
5	5	7	10	9	20	11	33
13	51	15	73	17	95	19	126
21	155	23	196	25	230	27	279
29	322	31	377	33	432	35	499
37	556	39	629	41	696	43	781
45	860	47	944	49	1023	51	1120
53	1211	55	1314	57	1425	59	1527
61	1656	63	1753	65	1873	67	2004
69	2125	71	2262				

Remark 8.3 (Geometric support versus polynomial divisibility)**.** *If $\overline{q}_n$ is squarefree, the failure of divisibility in* (23) *ensures at least one geometric root of $\overline{q}_n$ is absent from all predecessors over an algebraic closure. In the presence of repeated factors,* polynomial *nondivisibility might result solely from multiplicity; the theorem is stated at its actually certified strength. The universal open problem is exactly that C_n has no prime divisor greater than $2n$ for* every *odd n.*

9 Bernoulli anchors and primitive integer-coordinate ladders

Let B_k denote the ordinary Bernoulli numbers with $\mathsf{B}_2 = 1/6$, and let $Q_n = f_n$ denote the *monic* odd-order quotient, to distinguish it from the predecessor product $\mathcal{B}_n$.

Proposition 9.1 (Four Bernoulli-controlled anchor evaluations)**.** *For every odd $n \geq 5$,*

$$\begin{aligned} Q_n(0) &= \frac{2^{n-1}n}{n-1}\mathsf{B}_{n-1}, \quad Q_n(1) = (n-2)Q_n(0), \\ Q_n(2) &= \frac{n-1}{2}Q_n(0), \quad Q_n(3) = -\frac{(n-1)(n-2)}{6}Q_n(0). \end{aligned} \tag{24}$$

If $p > 2n$ divides the numerator of B_{n-1}, then $0, 1, 2, 3$ are four pairwise distinct roots of $\overline{Q}_n$.

Proof. Put $A(t) = t/(1 - e^{-t})$ and $L(t) = \log A(t)$. The identities

$$A(t) - A(-t) = t, \quad L(t) - L(-t) = t, \quad tA'(t) = (1 + t)A(t) - A(t)^2$$

permit extraction of the odd coefficients of $L(t)^2/2$, $A(t)L(t)$, $A(t)^2$, and $A(t)^3$. In particular $[X^2]S_n = [t^n]L^2/2 = -\mathsf{B}_{n-1}/(2(n-1)(n-1)!)$, which after $S_n = X^2(X-1)Q_n/(2^n n!)$ gives the first identity. Evaluating the corresponding derivative at $X = 1$ and the evaluations at $X = 2, 3$ gives the remaining rational multiples in (24). The denominators are p-units when $p > 2n$. □

The four residues are *target* roots, not necessarily new ones. Indeed $0, 1$ are already present in early columns, and 2 occurs in the immediate predecessor for the same Bernoulli irregularity. A genuine *primitive* residue at 3 requires the additional absence of zeros in every preceding column.

Proposition 9.2 (Exact first occurrence at coordinate 3)**.** *For $j \geq 5$ odd and $j \geq 4$ even, respectively,*

$$\begin{aligned} S_j(3) &= -\frac{3(j-2)}{2(j-1)!}\mathsf{B}_{j-1}, \\ S_j(3) &= \frac{j-1}{2j!}((j-2)\mathsf{B}_j + 2j\mathsf{B}_{j-2}). \end{aligned} \tag{25}$$

The initial values are $S_1(3) = 3/2$, $S_2(3) = 1$, $S_3(3) = 3/8$. At a good irregular-index pair (n, p), the coordinate 3 first occurs at n precisely when the odd Bernoulli values and the even Bernoulli linear combinations on the right of (25) *are all nonzero modulo p at prior orders.*

Proof. Iterate $tA' = (1+t)A - A^2$ to express A^3 as a combination of A, A', A'' with polynomial coefficients in t, then extract coefficients of each parity. In the good-prime range all displayed factorial and Bernoulli denominators are p-units, proving the necessary-and-sufficient first-occurrence criterion. □

Theorem 9.3 (Integer-coordinate Bernoulli staircase)**.** *For all positive integers r, n,*

$$\boxed{S_n(r+1) = \frac{r-n}{r}S_n(r) + S_{n-1}(r).} \tag{26}$$

If $n \geq 5$ is odd, $p > 2n$, and $h \geq 0$ satisfies $2h+3 \leq n-2$, then

$$\begin{aligned} &S_n(2) \equiv S_n(3) \equiv \cdots \equiv S_n(2h+3) \equiv 0 \pmod{p} \\ &\quad\Longleftrightarrow\quad \mathsf{B}_{n-1} \equiv \mathsf{B}_{n-3} \equiv \cdots \equiv \mathsf{B}_{n-2h-1} \equiv 0 \pmod{p}. \end{aligned} \tag{27}$$

In that case $\prod_{a=0}^{2h+3}(X-a)$ divides $\overline{Q}_n(X)$.

Proof. Multiply the differential identity for A by rA^{r-1} to obtain $A^{r+1} = (1+t)A^r - (t/r)(A^r)'$ and take the t^n-coefficient, giving (26). Write $S_n(r)$ as a linear combination $\sum_{j=0}^{r-1} c_{r,j}(n)S_{n-j}(1)$; induction from (26) gives $c_{r,r-1}(n) = 1$. Since n is odd and $S_k(1) = \mathsf{B}_k/k!$ for even $k \geq 2$, whereas $S_k(1) = 0$ for odd $k \geq 3$, the equations for $S_n(2), S_n(4), \ldots, S_n(2h+2)$ form a triangular system in the ordered Bernoulli values $\mathsf{B}_{n-1}, \mathsf{B}_{n-3}, \ldots, \mathsf{B}_{n-2h-1}$ with unit diagonal after dividing by factorials. The intervening odd integer evaluations involve only the same Bernoulli values and therefore vanish simultaneously. The factors $a^2(a-1)/(2^n n!)$ are units at the integer coordinates $2 \leq a \leq 2h+3$, and (24) provides the roots 0 and 1. This proves the factor divisibility. □

Example 9.4 (An uncovered simple 587-adic disk beyond the finite content range)**.** *For $(n, p) = (93, 587)$, exact arithmetic gives $\mathsf{B}_{90} \equiv \mathsf{B}_{92} \equiv 0 \pmod{587}$. Theorem 9.3 produces the six target roots $0, 1, 2, 3, 4, 5$ modulo 587. Yet the certificate checks*

$$\begin{aligned} &S_{93}(5) = 0, \quad S_{92}(5) = 179, \quad S'_{93}(5) = 498 \pmod{587}, \\ &S_m(5) \not\equiv 0 \pmod{587} \qquad (1 \leq m < 93). \end{aligned}$$

Therefore the root residue 5 is new. Since the derivative is nonzero, Hensel's lemma gives for every common 587-integral stationary source vector *a unique root of F_{93} inside $5 + 587\mathbb{Z}_{587}$, with none of the earlier columns vanishing in that disk. A double Bernoulli zero thus produces a* primitive *target root even though coordinate 3 is already present at orders 91 and 92.*

Remark 9.5 (A singular residue cannot be lifted automatically)**.** *At $(n, p, a) = (91, 277, -2)$ one has $S_{91}(-2) = S'_{91}(-2) = 0$ modulo 277, while $S''_{91}(-2) = 125 \not\equiv 0$. With zero sources, $F_{91}(-2) \equiv 277 \cdot 24 \pmod{277^2}$, so no root exists in the entire disk $-2 + 277\mathbb{Z}_{277}$ modulo 277^2. This does not contradict the mod-p primitive-root theorem; the elementary simple-root Hensel hypothesis is genuinely necessary.*

10 Proof provenance, scope and the uniform open problem

The proofs in this paper have distinct levels of logical dependence. The exact shift-companion identity and the Bernoulli staircase are direct symbolic identities; the all-order rational noncoverage theorem applies the *published* Feit–Roitman large-Zsigmondy theorem; the long-orbit exclusions of degrees $d-2, d-3$ use exact symbolic Euclidean and resultant identities in the order variable plus finite exceptional-prime checks. The content statement through $n = 71$ is a finite exact computer-assisted certificate; it does not imply its continuation to higher orders. Source-universal assertions rest only on coefficientwise $F_j \equiv S_j \pmod p$ for $p > 2n$ and on standard Hensel lifting when the necessary separability holds.

Every odd $n \geq 5$ has a nonzero rational remainder H_n by Theorem 6.3; whether its integer content has a prime divisor greater than $2n$ is not established. We therefore retain the following open problem.

Open Problem 10.1 (Uniform predecessor noncoverage)**.**

$$\boxed{\textit{For all odd } n \geq 5 \textit{ and every prime } p > 2n, \quad \overline{q}_n \nmid \overline{\mathcal{B}}_n \textit{ in } \mathbb{F}_p[X].} \tag{28}$$

Equivalently, C_n has no prime factor greater than $2n$ at every odd order. The claim has been proved in this manuscript only for $n \leq 71$ and for the explicit infinite families and special prime patterns covered by the preceding theorems.

The open problem must not be replaced by the stronger assertion that every geometric target root is absent from every preceding column: actual common roots occur, and in nonsquarefree characteristic polynomial nondivisibility need not detect an entirely new root position. A separate physical-realizability theorem for deterministic pushdown automata is likewise not implied by the formal stationary source setting. The CCR interpretation is conditional on the algebraic model (2) and integral source choice.

10.1 Finite certificate scope and verification

The results above have different forms of justification. The shift-companion and Bernoulli-ladder identities are symbolic derivations; the rational all-order noncoverage theorem relies on the published Feit–Roitman input and the explicit exceptional-order witnesses. The long-orbit degree-$(d-2)$ and degree-$(d-3)$ exclusions require the exact Euclidean identities in the order parameter. Their first two obstruction resultants are given in the proof, while the higher Euclidean coefficient and exceptional-residue checks are finite symbolic certificate components. The degree-$(d-2)$ argument uses the corrected linear-quotient chain in the $m = n-1$ case, not the superseded quadratic-quotient calculation.

The finite predecessor-content theorem uses exact rational coefficient arithmetic, primitive-part normalization, successive polynomial remaindering, denominator clearing and an integer coefficient gcd. All 34 exponents are stated in Table 3; they are not sampled statistics. An independently implemented Bernoulli recurrence and remainder calculation has reproduced the first 29 table rows, through odd order 61. The five remaining rows (63 through 71) are retained on the basis of the original complete finite certificate; their full high-order coefficient computations have not been repeated as part of this editorial audit. Small-order self-shift resultant checks, Feit-exception witnesses, and the explicit $n = 93$, $p = 587$ and $n = 91$, $p = 277$ residue calculations provide separate consistency checks, not substitutes for those exhaustive steps.

No executable source, verification script or supplementary data archive is part of, or promised with, this paper. The displayed formulae and certified finite outcomes carry the stated mathematical claims; for the heavy symbolic and full-content calculations, independent

end-to-end external checking remains an appropriate publication review step. We make no assertion of first-in-world priority for the classical ingredients or of physical-machine realizability from the formal stationary CCR model.

Statements and Declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, software, validation, writing—original draft, and writing—review and editing.
Funding. No external funding was received for this research.
Competing interests. The author declares no relevant financial or non-financial competing interests.
Data availability. This mathematical study uses no external empirical or personal dataset. The definitions, theorem statements, explicit arithmetic examples and scopes of the exact finite certificates are contained in the article. No supplementary code or data archive is provided as part of this publication.
Ethics approval and consent to participate. Not applicable.
Consent for publication. Not applicable.

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